

SEQ lane progress note

Route-1 / Posix_Card_Route1_Seq, private heap build green

The invariant being paid is not a collapsed-continuation recurrence. The target carrier for the lane is the head contribution

$$\text{card} \left((\text{single_root}(\text{seq}(h, t), k) \cup \text{single_term}(h, \sigma_4(t, k))) - (\mathcal{B}k \cup \mathcal{B}(\sigma_4(t, k))) \right) \leq \text{rsize}(h). \quad (\text{SEQ})$$

The useful accounting charges the head size $\text{rsize}(h)$. Charging $D_1(h, \sigma_4(t, k))$ after the continuation has already collapsed is the false recurrence: it loses the extra root row that appears in the `seq`-head case.

The RCHAR case is now a one-row proof. The term component is exactly the new boundary,

$$\text{single_term}(\text{char}(c), \sigma_4(t, k)) = \mathcal{B}(\sigma_4(t, k)),$$

so it disappears after subtracting $\mathcal{B}k \cup \mathcal{B}(\sigma_4(t, k))$. What remains is the root row, and the checked lemma is

$$\text{card}(\text{single_root}(\text{seq}(\text{char}(c), t), k)) \leq 1. \quad (\text{RCHAR-root})$$

Thus the branch closes immediately:

$$\begin{aligned} & \text{card} \left((\text{single_root}(\text{seq}(\text{char}(c), t), k) \cup \text{single_term}(\text{char}(c), \sigma_4(t, k))) - (\mathcal{B}k \cup \mathcal{B}(\sigma_4(t, k))) \right) \\ & \leq \text{rsize}(\text{char}(c)). \end{aligned}$$

This was committed as the RCHAR root helper and the RCHAR head-core case.

The star boundary is a one-step simulation, but only after subtraction. The tempting statement

$$\text{card}(\mathcal{B}(\sigma_4(\text{star}(r), k))) \leq 1$$

is false: when the star collapses to 1, the whole old boundary $\mathcal{B}k$ can reappear. The checked statement is the differential one:

$$\text{card}(\mathcal{B}(\sigma_4(\text{star}(r), k)) - \mathcal{B}k) \leq 1. \quad (\text{STAR-boundary})$$

The proof first shows the row-level form

$$\text{card} \left(\text{row}(S(\sigma_4(\text{star}(r), k))) - \text{row}(S(k)) \right) \leq 1,$$

then uses the verified inclusion $\text{row}(S(k)) \subseteq \mathcal{B}k$. The root identity also holds:

$$\text{single_root}(\text{star}(r), k) = \mathcal{B}(\sigma_4(\text{star}(r), k)). \quad (\text{STAR-root})$$

The singleton-size ledger now has its star and sequence assembly bricks. For singleton carriers

$$D_1(q, k) = \text{card}(\mathcal{A}(\text{alts}[q], k) - \mathcal{B}k),$$

the atom leaves are green:

$$D_1(0, k) \leq \text{rsize}(0), \quad D_1(\text{char}(c), k) \leq \text{rsize}(\text{char}(c)).$$

The star recurrence is now checked:

$$D_1(\text{star}(r), k) \leq 1 + D_1(r, \sigma_4(\text{star}(r), k)). \quad (\text{D1-STAR})$$

Hence any child bound $D_1(r, \sigma_4(\text{star}(r), k)) \leq \text{rsize}(r)$ immediately gives

$$D_1(\text{star}(r), k) \leq \text{rsize}(\text{star}(r)).$$

For a sequence head, the set algebra has also been isolated. Let

$$c = \sigma_4(r_2, k).$$

If the SEQ-head core for r_1 is bounded by $\text{rsize}(r_1)$, and if the BND lane supplies the boundary absorption

$$\text{card}((\mathcal{B}c - \mathcal{B}k) \cup (\text{single_term}(r_2, k) - \mathcal{B}k)) \leq D_1(r_2, k),$$

then the checked assembly lemma gives

$$D_1(\text{seq}(r_1, r_2), k) \leq \text{rsize}(r_1) + D_1(r_2, k). \quad (\text{D1-SEQ})$$

What remains is exactly the root-slack and cover work. The unclosed part is not another atom calculation. It is the structural root row in the seq-head case, plus the alternation branch cover:

$$\begin{aligned} \sigma_4(\text{seq}(r_1, r_2), k) &= \sigma_4(r_1, \sigma_4(r_2, k)), \\ \text{single_root}(\text{seq}(\text{seq}(r_1, r_2), t), k) &\not\equiv \text{single_root}(\text{seq}(r_1, \text{seq}(r_2, t)), k) \end{aligned}$$

definitionally. That extra row is the genuine +1 paid by $\text{rsize}(\text{seq}(r_1, r_2)) = 1 + \text{rsize}(r_1) + \text{rsize}(r_2)$, not a row to be erased by reassociation.

The remaining imported dependencies should be stated explicitly when the main lemma is assembled:

BND: `boundary_term_absorb`, COVER: L1 singleton cover.

The current file is green under the private build command from `ROUTE_SEQ.md`; no shared heap wrapper was used.

Last green commits: `0511e87` (RSEQ D1 assembly), `ff7c8b6` (D1 star corollary), `39e4fba` (D1 star step), `62ca737` (star boundary shift), `fbface5/6eff89b` (RCHAR core/root).